

# Initial Model Derivation from Migrated Appendix

Migrated Appendix A–B source material, organized for repository review

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This file is the source of truth for long mathematical derivations. Markdown files should only summarize it.

**Provenance note.** This document mechanically copies Appendix A, “Model Construction,” and Appendix B, “Nondimensionalization,” from `paper/appendix/appendix.tex`. The copied source begins at the Model Construction section and stops before the Homogeneous Diagnostic Derivation section. Wrapper packages and macro definitions were added only so this derivation can compile as a standalone document. No scientific corrections, equation edits, or validation claims are made here.

## 1 Model Construction

We formulate a one-dimensional Lagrangian model for the coupled thermo-chemo-poroelastic dynamics of a thermoresponsive gel slab undergoing an exothermic reaction. The conservative variables are the local swelling ratio  $J(X, t)$ , the reactant content per reference volume  $Jc$ , and the temperature  $T(X, t)$ ; the solvent chemical potential is used as an algebraic mixed variable.

### 1.1 Kinematics

Consider a gel slab of reference (as-prepared) half-thickness  $H_0$ . We adopt a material (Lagrangian) coordinate  $X \in [0, H_0]$ , with  $X = 0$  denoting the symmetry plane and  $X = H_0$  the free surface. The current position of a material point is  $x(X, t)$ , and the local volume ratio is

$$J(X, t) = \frac{\partial x}{\partial X} > 0. \quad (1)$$

In one dimension,  $J$  is the local swelling ratio. The polymer volume fraction is a dependent variable determined by mass conservation of the network:

$$\phi(X, t) = \frac{\phi_{p0}}{J(X, t)}, \quad (2)$$

where  $\phi_{p0}$  is the polymer volume fraction in the reference (as-prepared) state;  $\phi_{p0} = 0.15$  corresponds to a gel that is 85% solvent by volume at preparation. The current thickness of the slab is

$$h(t) = x(H_0, t) = \int_0^{H_0} J(X, t) dX. \quad (3)$$

## 1.2 Governing equations

### 1.2.1 Swelling–permeation equation

The evolution of  $J$  is governed by solvent conservation:

$$\frac{\partial J}{\partial t} = -\frac{\partial Q_s}{\partial X}, \quad (4)$$

where  $Q_s$  is the solvent flux relative to the polymer network, driven by chemical-potential gradients:

$$Q_s = -M_s(J, T) \frac{\partial \mu_s}{\partial X}. \quad (5)$$

Here  $M_s(J, T)$  is the permeation mobility and  $\mu_s$  is the solvent chemical potential.

### 1.2.2 Reactant transport equation

The reactant concentration  $c(X, t)$  (moles per unit current total volume) evolves according to

$$\frac{\partial(Jc)}{\partial t} + \frac{\partial N_c}{\partial X} = -J R(c, T, J), \quad (6)$$

with the total reactant flux

$$N_c = c Q_s - D_c(J, T) \frac{\partial c}{\partial X}. \quad (7)$$

The first term represents advective transport by the solvent flow, and the second term Fickian diffusion within the pore fluid. The effective diffusivity  $D_c(J, T)$  depends on the local microstructure via  $J$ .

### 1.2.3 Heat equation

The temperature field satisfies

$$C_{\text{eff}}(J) \frac{\partial T}{\partial t} = \frac{\partial}{\partial X} \left( K_T(J) \frac{\partial T}{\partial X} \right) - \frac{\partial}{\partial X} \left( c_p^\ell T Q_s \right) + (-\Delta H_r) J R(c, T, J), \quad (8)$$

where  $C_{\text{eff}}(J)$  is the effective volumetric heat capacity,  $K_T(J)$  the effective thermal conductivity,  $c_p^\ell$  the solvent heat capacity per unit volume, and  $-\Delta H_r > 0$  the exothermic reaction enthalpy. The three right-hand-side terms represent thermal conduction, enthalpy advection by the solvent flow, and reaction heat generation, respectively.

## 1.3 Free energy and chemical potential

The Helmholtz free-energy density per unit reference volume is written as

$$\Psi(J, T) = \Psi_{\text{el}}(J) + \Psi_{\text{mix}}(\phi, T) + \frac{\kappa_J}{2} \left( \frac{\partial J}{\partial X} \right)^2, \quad (9)$$

where  $\phi = \phi_{p0}/J$ , and the three contributions are the elastic energy of the network, the Flory–Huggins mixing energy, and a gradient-energy regularization.

For the polymer network we adopt a neo-Hookean elastic energy appropriate for uniaxial swelling in the slab geometry:

$$\Psi_{\text{el}}(J) = \frac{G}{2}(J^2 - 1 - 2 \ln J), \quad (10)$$

where  $G$  is the shear modulus of the reference-state network.

The mixing free-energy density takes the standard Flory–Huggins form [?]:

$$\Psi_{\text{mix}}(\phi, T) = \frac{R_g T}{v_s} \left[ (1 - \phi) \ln(1 - \phi) + \chi(T, \phi) \phi(1 - \phi) \right], \quad (11)$$

with  $R_g$  the gas constant,  $v_s$  the solvent molar volume, and the Flory–Huggins interaction parameter

$$\chi(T, \phi) = \chi_0(T) + \chi_1 \phi. \quad (12)$$

The temperature-dependent part  $\chi_0(T)$  encodes the LCST behavior of the gel;  $\chi_1$  captures the leading concentration dependence.

The solvent chemical potential is obtained as  $\mu_s = v_s \partial \Psi / \partial(1 - \phi) \cdot \partial(1 - \phi) / \partial J$  [?]. Evaluating this derivative yields the mixing contribution

$$\frac{\mu_{\text{mix}}}{R_g T / v_s} = \ln(1 - \phi) + \phi + \chi(T, \phi) \phi^2 - \phi^2(1 - \phi) \frac{\partial \chi}{\partial \phi}, \quad (13)$$

so that the linear form (12) gives  $\partial \chi / \partial \phi = \chi_1$ . The shorter expression without this final term is retained only as an effective-osmotic comparison closure, not as a strict free-energy derivative. and the elastic contribution

$$\frac{\mu_{\text{el}}}{R_g T / v_s} = \frac{G v_s}{R_g T} (J - J^{-1}). \quad (14)$$

The full chemical potential, including the gradient-energy regularization, is

$$\mu_s = \mu_{\text{mix}}(J, T) + \mu_{\text{el}}(J) - \kappa_J \frac{\partial^2 J}{\partial X^2}. \quad (15)$$

## 1.4 Reaction kinetics

The volumetric reaction rate takes the Arrhenius form modulated by a collapse-dependent accessibility factor:

$$R(c, T, J) = k_0 c^n \exp\left(-\frac{E_a}{R_g T}\right) a(J), \quad (16)$$

where  $k_0$  is the pre-exponential factor,  $n$  the reaction order, and  $E_a$  the activation energy.

The activity factor  $a(J)$  models the suppression of reaction due to reduced free solvent volume upon collapse:

$$a(J) = A_{\text{min}} + (1 - A_{\text{min}}) \bar{p}^{m_{\text{act}}}, \quad \bar{p} = \text{clip}\left[\frac{1 - \phi}{1 - \phi_{\text{ref}}}, 0, 1\right]. \quad (17)$$

When the gel collapses ( $J \rightarrow \phi_{p0}$ ,  $\phi \rightarrow 1$ ), the activity approaches the residual floor  $A_{\text{min}}$ . This is a phenomenological porosity/accessibility closure rather than a microscopic derivation.

The effective transport coefficients similarly depend on the microstructure:

$$D_c(J) = D_0 [D_{\text{min}} + (1 - D_{\text{min}}) \bar{p}^{m_{\text{diff}}}], \quad (18)$$

$$M_s(J) = M_0 \bar{p}^{m_{\text{mob}}}, \quad (19)$$

with optional Mackie–Meares and free-volume forms used only for comparison. Thus solute diffusion and solvent permeation are suppressed but not forced to vanish in the collapsed state.

## 1.5 Boundary conditions

At the symmetry plane  $X = 0$ :

$$Q_s = 0, \quad N_c = 0, \quad \frac{\partial T}{\partial X} = 0. \quad (20)$$

At the free surface  $X = H_0$ :

**Solvent exchange.** A Robin-type condition couples the surface chemical potential to the bath:

$$Q_s = L_\mu(\mu_s - \mu_{\text{bath}}), \quad (21)$$

where  $L_\mu$  is a surface transfer coefficient. In the limit of rapid equilibration,  $\mu_s = \mu_{\text{bath}}$ .

**Reactant supply.**

$$N_c = k_c(c - c_b), \quad (22)$$

where  $c_b$  is the constant bath concentration.

**Heat exchange.**

$$-K_T \frac{\partial T}{\partial X} = h_T(T - T_\infty), \quad (23)$$

where  $h_T$  is the surface heat-transfer coefficient and  $T_\infty$  the ambient temperature.

**Mechanical boundary.** The free surface is traction-free. In the  $J$ -formulation this is implicitly satisfied by the chemical-potential balance.

## 2 Nondimensionalization

### 2.1 Scaling

We introduce the following dimensionless variables. Length is scaled by the reference thickness:

$$X = H_0 \tilde{x}, \quad \tilde{x} \in [0, 1]. \quad (24)$$

The swelling ratio  $J$  is already dimensionless. Reactant concentration is normalized by the bath value:

$$c = c_b u. \quad (25)$$

Temperature is measured relative to ambient, scaled by the adiabatic reaction temperature rise:

$$T = T_\infty + \Delta T_* \theta, \quad \Delta T_* = \frac{(-\Delta H_r) c_b}{C_*}, \quad (26)$$

where  $C_*$  is a representative volumetric heat capacity.

The chemical potential is scaled by the thermal mixing energy:

$$\mu_s = \mu_* m, \quad \mu_* = \frac{R_g T_\infty}{v_s}. \quad (27)$$

Time is scaled by the swelling/permeation time:

$$\tau_s = \frac{H_0^2}{M_0 \mu_*}, \quad (28)$$

where  $M_0$  is a reference mobility. The solvent and reactant flux scales follow:

$$Q_s = \frac{H_0}{\tau_s} q, \quad N_c = \frac{c_b H_0}{\tau_s} n. \quad (29)$$

Material functions are factored as reference value times dimensionless function:

$$M_s = M_0 \mathcal{M}(J, \theta), \quad D_c = D_0 \mathcal{D}_{\text{cur}}(J, \theta), \quad (30)$$

$$C_{\text{eff}} = C_* \mathcal{C}(J), \quad K_T = K_* \mathcal{K}_{\text{cur}}(J). \quad (31)$$

In the reference-coordinate equations below, the baseline closure uses effective reference-coordinate coefficients  $\mathcal{D}_{\text{ref}}$ ,  $\mathcal{K}_{\text{ref}}$ , and  $\mathcal{C}_{\text{ref}}$ . When the tabulated coefficients are interpreted as current-space material properties, the metric-corrected comparison uses  $\mathcal{D}_{\text{ref}} = \mathcal{D}_{\text{cur}}/J$ ,  $\mathcal{K}_{\text{ref}} = \mathcal{K}_{\text{cur}}/J$ , and  $\mathcal{C}_{\text{ref}} = J\mathcal{C}_{\text{cur}}$ .

## 2.2 Dimensionless reaction rate

Defining the reference reaction rate

$$R_0 = k_0 c_b^n \exp\left(-\frac{E_a}{R_g T_\infty}\right), \quad (32)$$

we write  $R = R_0 \hat{R}(u, \theta, J)$  with

$$\hat{R}(u, \theta, J) = u^n a(J) \exp\left[\frac{\Gamma_A \theta}{1 + \varepsilon_T \theta}\right], \quad (33)$$

where

$$\varepsilon_T = \frac{\Delta T_*}{T_\infty}, \quad \Gamma_A = \frac{E_a \Delta T_*}{R_g T_\infty^2}. \quad (34)$$

When  $\varepsilon_T \ll 1$ , this simplifies to  $\hat{R} \approx u^n a(J) e^{\Gamma_A \theta}$ .

## 2.3 Dimensionless chemical potential

The dimensionless chemical potential is

$$m = m_{\text{mix}}(J, \theta) + m_{\text{el}}(J) - \ell^2 \frac{\partial^2 J}{\partial \tilde{x}^2}, \quad (35)$$

with the interface parameter

$$\ell^2 = \frac{\kappa_J}{\mu_* H_0^2}. \quad (36)$$

The mixing part retains the Flory–Huggins form (neglecting the  $T/T_\infty \approx 1 + \varepsilon_T \theta \approx 1$  prefactor, valid when  $\varepsilon_T \ll 1$ ):

$$m_{\text{mix}} = \ln(1 - \phi) + \phi + \chi(\theta, \phi) \phi^2 - \phi^2(1 - \phi) \frac{\partial \chi}{\partial \phi}, \quad \phi = \frac{\phi_{p0}}{J}, \quad (37)$$

with the dimensionless interaction parameter

$$\chi(\theta, \phi) = \chi_\infty + S_\chi \theta + \chi_1 \phi, \quad (38)$$

where  $\chi_\infty = \chi_0(T_\infty)$  and the thermo-swelling sensitivity is

$$S_\chi = \Delta T_* \left. \frac{d\chi_0}{dT} \right|_{T_\infty}. \quad (39)$$

For the linear concentration dependence in Eq. (38),  $\partial\chi/\partial\phi = \chi_1$ .

The elastic contribution follows from Eq. (14):

$$m_{\text{el}} = \Omega_e (J - J^{-1}), \quad \Omega_e = \frac{G v_s}{R_g T_\infty}, \quad (40)$$

measuring the network stiffness relative to the thermal mixing energy.

## 2.4 Dimensionless governing equations

Dropping tildes for clarity, the dimensionless system reads:

**Swelling equation.**

$$\frac{\partial J}{\partial t} = \frac{\partial}{\partial x} \left( \mathcal{M}(J, \theta) \frac{\partial m}{\partial x} \right), \quad (41)$$

**Chemical potential.**

$$m = m_{\text{mix}}(J, \theta) + m_{\text{el}}(J) - \ell^2 \frac{\partial^2 J}{\partial x^2}, \quad (42)$$

**Reactant equation.**

$$\frac{\partial(Ju)}{\partial t} + \frac{\partial}{\partial x} \left( u q - \delta \mathcal{D}_{\text{ref}}(J, \theta) \frac{\partial u}{\partial x} \right) = -\text{Da} J \hat{R}(u, \theta, J), \quad (43)$$

where  $q = -\mathcal{M}(J, \theta) \partial_x m$  and

$$\delta = \frac{D_0}{M_0 \mu_*} \quad (44)$$

is the ratio of solute diffusion speed to solvent permeation speed.

The Damköhler number, measuring reaction rate relative to swelling rate, is

$$\text{Da} = \tau_s k_0 c_b^{n-1} \exp\left(-\frac{E_a}{R_g T_\infty}\right). \quad (45)$$

**Heat equation.**

$$\begin{aligned} c_{\text{ref}}(J) \frac{\partial \theta}{\partial t} = & \alpha \frac{\partial}{\partial x} \left( \mathcal{K}_{\text{ref}}(J) \frac{\partial \theta}{\partial x} \right) + \text{Da} J \hat{R}(u, \theta, J) \\ & - \alpha \text{Pe}_T \frac{\partial}{\partial x} [(\Theta_\infty + \theta) q], \end{aligned} \quad (46)$$

with

$$\alpha = \frac{K_* \tau_s}{C_* H_0^2} = \frac{\tau_s}{\tau_T}, \quad \text{Pe}_T = \frac{c_p^\ell M_0 \mu_*}{K_*}, \quad \Theta_\infty = \frac{T_\infty}{\Delta T_*}, \quad (47)$$

where  $\tau_T = C_* H_0^2 / K_*$  is the thermal diffusion time. When  $\text{Pe}_T \ll 1$ , the convective term may be dropped.

**Constraint.**

$$\phi = \frac{\phi_{p0}}{J}. \quad (48)$$

## 2.5 Dimensionless boundary conditions

At  $x = 0$  (symmetry):

$$q = 0, \quad u q - \delta \mathcal{D}_{\text{ref}} u_x = 0, \quad \theta_x = 0. \quad (49)$$

At  $x = 1$  (free surface):

$$q = \text{Bi}_\mu (m - m_b), \quad (50)$$

$$u q - \delta \mathcal{D}_{\text{ref}} u_x = \text{Bi}_c (u - 1), \quad (51)$$

$$-\alpha \mathcal{K}_{\text{ref}}(J) \theta_x = \text{Bi}_T \theta, \quad (52)$$

where the surface Biot numbers for permeation and mass transfer are

$$\text{Bi}_\mu = \frac{L_\mu H_0}{M_0}, \quad \text{Bi}_c = \frac{k_c H_0}{M_0 \mu_*}, \quad (53)$$

and the thermal exchange number is

$$\text{Bi}_T = \frac{h_T \tau_s}{C_* H_0} = \alpha \frac{h_T H_0}{K_*}, \quad (54)$$

which measures the surface heat loss rate directly on the swelling timescale  $\tau_s$ . The standard thermal Biot number  $h_T H_0 / K_* = \text{Bi}_T / \alpha$  is recovered by factoring out  $\alpha$ .

## 2.6 Controlling parameter groups

The oscillatory dynamics are governed by the following dimensionless groups:

- Da — Damköhler number: strength of reaction heat source relative to swelling.
- $\alpha = \tau_s / \tau_T$  — thermal diffusion relative to swelling.
- $S_\chi$  — thermo-swelling sensitivity: how strongly temperature shifts the interaction parameter.
- $\Gamma_A$  — Arrhenius feedback strength.
- $\delta$  — solute diffusion to permeation ratio.
- $\ell$  — interface regularization length.
- $\text{Bi}_c$  — surface reactant Biot number.
- $\text{Bi}_T$  — thermal exchange number (surface heat loss on the swelling timescale).

Together with the working-point parameters  $\phi_{p0}$  and  $\chi_\infty$ , these form the minimal set controlling the onset of self-oscillation. In physical terms, oscillation requires that the fast positive feedback (reaction  $\rightarrow$  heating  $\rightarrow$  collapse) and the slow negative feedback (collapse  $\rightarrow$  transport suppression  $\rightarrow$  cooling  $\rightarrow$  re-swelling) operate on well-separated time scales.